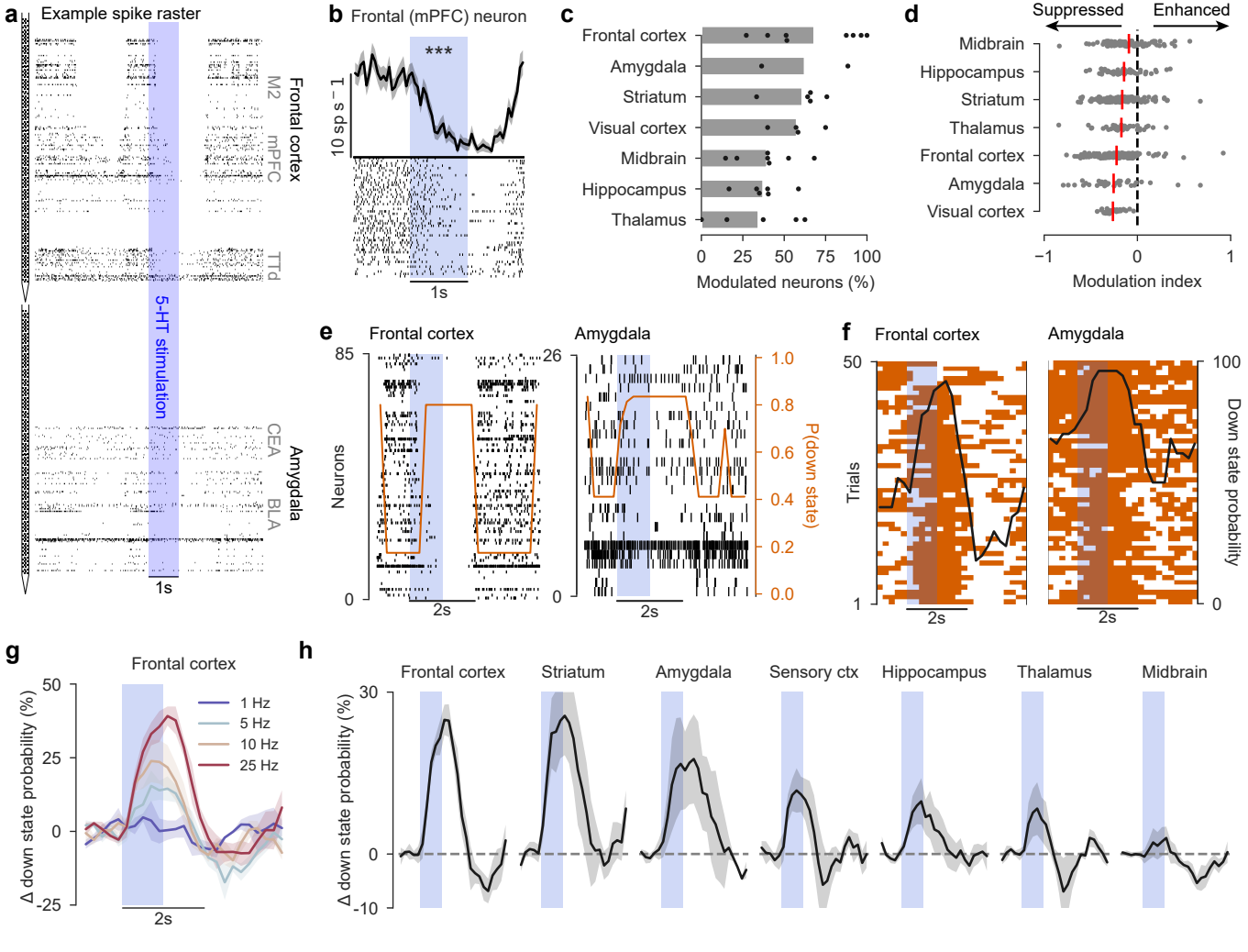

a
Example spike raster
Frontal cortex
M2
mPFC
TTd
5-HT stimulation
Amygdala
CEA
BLA
1s
b
Frontal (mPFC) neuron
***
10 sp s − 1
1s
c
Frontal cortex
Amygdala
Striatum
Visual cortex
Midbrain
Hippocampus
Thalamus
0
25
50
75
100
Modulated neurons (%)
d
Suppressed
Enhanced
Midbrain
Hippocampus
Striatum
Thalamus
Frontal cortex
Amygdala
Visual cortex
−1
0
1
Modulation index
e
Frontal cortex
Amygdala
85
0
Neurons
26
0
2s
2s
1.0
0.8
0.6
0.4
0.2
0.0
P(down state)
f
Frontal cortex
Amygdala
50
1
Trials
100
0
Down state probability
2s
2s
g
Frontal cortex
50
25
0
-25
Δ down state probability (%)
1 Hz
5 Hz
10 Hz
25 Hz
2s
h
Frontal cortex
Striatum
Amygdala
Sensory ctx
Hippocampus
Thalamus
Midbrain
30
0
-10
Δ down state probability (%)